

Deepanjay Nandal

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SUMMARY

Backend Software Engineer with 3+ years building production systems, from CVE validation infrastructure for Google's Android Security team to event-driven pipelines processing 100k+ daily events. M.S. from Arizona State University, with experience across Kafka/Redis pipelines, hybrid RAG search, and LLM inference caching. Experienced in scalable system design, low-latency API development, and building fault-tolerant distributed systems end-to-end.

EDUCATION

Arizona State University

Master of Science (M.S.) in Computer Software Engineering

Tempe, Arizona, USA

Aug 2024 – May 2026

Vellore Institute of Technology

Bachelor of Technology (B.Tech) in Information Technology

Tamil Nadu, India

Jul 2017 – Jul 2021

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, Node.js, SQL

Backend Technologies: Distributed Systems, Microservices, Event-Driven Architecture, Multi-tenancy, REST APIs, WebSockets, Message Queues, Rate Limiting, Socket.IO

Data & Infrastructure: Apache Kafka, PostgreSQL, PostGIS, Redis, pgvector, Distributed Caching, BullMQ

Security & Systems: Android Open Source Project (AOSP), Binary Analysis, Ghidra, Frida, ClusterFuzz

Cloud & DevOps: AWS (EC2, Lambda, SQS, IoT Greengrass, SES), GCP, Firebase, Cloudflare, Docker, CI/CD

AI/ML: Retrieval-Augmented Generation (RAG), Vector Search, Large Language Model (LLM) APIs, Semantic Caching, Embeddings, Collaborative Filtering, Retrieval Evaluation

EXPERIENCE

Discover Excellence LLC

Software Engineering Intern

California, USA

Jan 2026 – June 2026

- Built a high-throughput Kafka-based visitor intelligence pipeline for an AI-powered eCommerce analytics platform, processing 100k+ daily behavioral events at sub-40ms p95 latency to drive discount outreach via AWS SES.
- Designed in-memory Redis collaborative filtering for product affinity scoring, driving measurable lift in conversion.
- Served personalized product feeds at the edge via Cloudflare, achieving 92% cache hit rate and sub-30ms p95 latency for multi-tenant deployments.
- Implemented fault-tolerant BullMQ background pipeline for async job processing, handling 50k+ daily jobs with guaranteed job delivery through persistent state, retries, exponential backoff, and idempotency keys.

Ittiam Systems Pvt Ltd

Senior Software Engineer

Bengaluru, India

Mar 2022 – Jun 2024

- Authored 50+ CTS/STS validation tests merged into the Android Open Source Project (AOSP), each proving a CVE patch for Google's Android Security team, working directly with Google engineers.
- Developed a Ghidra-based binary analysis approach for the Android Bluetooth stack that Google had marked infeasible, which the team adopted to validate 40+ Bluetooth CVEs.
- Engineered a Frida instrumentation framework in Python and Java to hook private Android methods at runtime, enabling the team to reproduce 30+ CVEs that standard testing could not reach across multiple Android releases.
- Built GCP Python pipeline to ingest ClusterFuzz crash outputs across Dockerized environments, auto-classifying reports by CVSS severity into a PostgreSQL-backed schema, cutting manual triage overhead by 35%.
- Automated Google Gerrit CI/CD workflows via idempotent webhook-triggered Python scripts, replacing manual submission triggering and handling 30–40 CVE submissions per month.

PROJECTS

BitMod (AI Inference Cache) | *Python, FastAPI, Vector Embeddings, Multi-tenant, Docker*

- Architected a 9-layer LLM cache proxy with SHA-256 exact lookup, semantic reuse, and Bayesian evidence scoring, achieving up to 94% hit rate and 176x speedup at 71ms cached latency.
- Established multi-tenant namespace isolation, intent-based cache routing with full LLM bypass for deterministic queries, and a knowledge ingestion pipeline that eliminates cold start by pre-seeding 50k+ cached Q&A pairs.

WeVibe | *Appy.yo (Client Project) | Node.js, TypeScript, PostgreSQL, Socket.IO, Redis, Google Cloud Run*

- Created a real-time matchmaking backend using Socket.IO and Redis pub/sub for horizontally scalable event delivery, PostgreSQL advisory locks preventing race conditions, and PostGIS geo-filtering on Google Cloud Run.
- Integrated Gemini bio generation with a custom prompt-injection guard and rate limiting (5 bios/day, 60s cooldown), Firebase JWT auth, and Apple Sign-In with revocation, validated across 17 Jest integration suites.

CaseTally (Legal Search Platform) | *FastAPI, pgvector, RAG, BM25, LLM APIs*

- Constructed a hybrid legal search system over 33k+ U.S. Code chunks using BM25 and pgvector, with LLM query rewriting improving MRR by 21% and weighted score fusion across both retrieval paths.
- Built an idempotent ingestion pipeline across 54 U.S. Code titles with SHA256-based change detection and a FastAPI SSE streaming backend, with raw retrieval benchmarked at p50 6ms and max 22ms.